COMPARATIVE STUDY OF CUBIC EQUATIONS OF STATE

Real-Gas PVT Behaviour, Fugacity, Enthalpy and Entropy

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Abstract

Real-gas calculations become important when pressure is elevated, temperature approaches a phase boundary, or molecular interactions cause measurable departures from the ideal-gas law. This report brings together the thermodynamic framework and computational approaches described in two supplied project reports. Four cubic equations of state - van der Waals (vdW), Redlich-Kwong (RK), Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR) - are considered for estimating compressibility, molar volume, density, residual enthalpy and entropy, and fugacity. The source material describes two distinct implementations: a Python command-line program with a 22-gas property set and a browser calculator with nine gas options and an interactive P-V plot. Published examples from both reports are compared, while the provenance and validation status of each numerical result are kept explicit. The emphasis is on understanding how model choice affects predicted properties and on identifying checks required before engineering use.

1. Objective and scope

The objective is to examine how cubic equations of state translate critical-property data into practical real-gas property estimates, and to compare the computational design and reported performance of the two reference implementations. The present report is a comparative synthesis: it does not claim that either supplied software package was re-coded or independently executed for this submission.

- Summarize the common cubic-EOS formulation and the parameters that distinguish the four models.
- Explain the calculation route for compressibility roots, volume, density, residual H and S, fugacity, and saturation pressure.
- Compare the Python and browser-based program structures using only features stated in the source reports.
- Interpret the available n-butane and carbon-dioxide examples and separate benchmark comparisons from illustrative, unvalidated outputs.

2. Thermodynamic model

A real gas is represented through a pressure-volume-temperature relation that accounts approximately for finite molecular size and attractive interactions. The shared generalized form used in the first report is:

$$P = RT/(V - b) - a(T)/[(V + \varepsilon b)(V + \sigma b)]$$

Here, V is molar volume, R is the gas constant, T is absolute temperature, b represents the co-volume correction, and a(T) is the temperature-dependent attraction parameter. The constants Ω and Ψ scale b and a using critical temperature Tc and critical pressure Pc; α is the reduced-temperature correction and ω is the acentric factor.

$$a(T) = \Psi\, \alpha(Tr, \omega)\, R^2Tc^2/Pc \quad ; \quad b = \Omega\, RTc/Pc \quad ; \quad Tr = T/Tc$$

The parameter sets reported for the four models are compared below. These constants are retained as presented in the first source report [1, pp. 1-2].

EOS	α(Tr)	σ	ε	Ω	Ψ	Zc
vdW	1	0	0	1/8	27/64	3/8
RK	Tr^(-1/2)	1	0	0.08664	0.42748	1/3
SRK	[1 + κ(1 - √Tr)]²; κ = 0.480 + 1.574ω - 0.176ω²	1	0	0.08664	0.42748	1/3
PR	[1 + κ(1 - √Tr)]²; κ = 0.37464 + 1.54226ω - 0.26992ω²	1 + √2	1 - √2	0.07780	0.45724	0.30740

Notation note: some lecture-note transcriptions contain coefficient slips. The first report explicitly states that its SRK and PR κ expressions use the published coefficients shown above [1, p. 6].

2.1 Compressibility equation and phase roots

The compressibility factor provides a dimensionless measure of deviation from ideal-gas behaviour. For one mole, it connects the EOS solution to molar volume:

$$Z = PV/(RT) \quad ; \quad V = ZRT/P \quad ; \quad \rho = M/V$$

Rearranging the generalized cubic EOS in terms of Z produces a third-degree polynomial. With $A = aP/(RT)^2$, $B = bP/(RT)$, $u = \sigma + \varepsilon$, and $w = \sigma\varepsilon$, the form reported in the Python project is:

$$Z^3 + c_2Z^2 + c_1Z + c_0 = 0$$

$$c_2 = -1 - B + uB \quad ; \quad c_1 = A + wB^2 - uB - uB^2 \quad ; \quad c_0 = -AB - wB^2 - wB^3$$

When three real roots are obtained, the smallest and largest roots are associated with liquid-like and vapour-like states, respectively; the intermediate root is not selected as a stable phase. The Python report describes solving the largest root with Newton-Raphson, reducing the cubic to a quadratic for the remaining roots, and retaining roots satisfying $Z > B$. It also describes a thermodynamic stability choice based on the lower residual Gibbs-energy value, while allowing the user to force a vapour or liquid root for metastable-state studies [1, pp. 2-3].

2.2 Residual enthalpy and entropy

The non-ideal contribution is evaluated relative to the ideal-gas reference at the same state. For the generalized cubic form, the first report defines an auxiliary logarithmic term I and gives closed-form residual-property relations:

$$I = (1/(\sigma - \varepsilon)) \ln[(Z + \sigma B)/(Z + \varepsilon B)] \qquad (\text{vdW: } I = B/Z)$$

$$H^R/(RT) = Z - 1 + [d(\ln \alpha)/d(\ln Tr) - 1]\, qI$$

$$S^R/R = \ln(Z - B) + [d(\ln \alpha)/d(\ln Tr)]\, qI \quad ; \quad q = A/B$$

Total property changes combine the ideal-gas heat-capacity contribution with the difference in residual properties between the final and initial states. The heat-capacity correlation in the Python report is $Cp,ig/R = A + BT + CT^2 + DT^{-2}$:

$$\Delta H = R \int (Cp,ig/R)dT + H_2^R - H_1^R$$

$$\Delta S = R \int (Cp,ig/R)(dT/T) - R \ln(P_2/P_1) + S_2^R - S_1^R$$

The reported absolute H and S reference is an ideal gas at 298.15 K and 1 bar, where both properties are assigned zero [1, p. 2; 2, p. 2].

2.3 Fugacity and phase equilibrium

Fugacity replaces pressure in the chemical-potential expression for a non-ideal gas. For a pure component, it is related to pressure through the fugacity coefficient:

$$f = \varphi P \quad ; \quad \ln \varphi = Z - 1 - \ln(Z - B) - qI$$

At vapour-liquid equilibrium, the fugacities of the coexisting liquid and vapour phases are equal. The browser calculator searches for a saturation pressure at which the two phase fugacity coefficients match. The Python report describes successive substitution, updating pressure by $P \leftarrow P(\varphi L/\varphi V)$, until the relative change is sufficiently small. For a compressed liquid, the Python report also gives a Poynting correction using saturated-liquid fugacity and liquid molar volume [1, pp. 2, 4].

$$fL = Psat\ \varphi sat \exp[VL(P - Psat)/(RT)]$$

Applicability: a saturation-pressure estimate requires a subcritical temperature and suitable liquid-like and vapour-like roots. A numerical search may fail when the required roots are absent or the selected search interval does not bracket a solution [2, p. 4].

3. Computational approaches and program workflow

The two source reports describe different software forms. They share the same broad thermodynamic task but should not be treated as one combined application: one is a text-driven Python calculation tool, whereas the other is a browser interface with interactive plotting.

Feature	Python implementation [1]	Browser implementation [2]
Application form	Command-line program: cubic_eos.py	Browser-based calculator using HTML, CSS and JavaScript
Gas property set	22 gases with critical constants and ideal-gas Cp coefficients	Nine gases: methane, ethane, propane, butane, nitrogen, oxygen, CO_2, ammonia and water
Main outputs	PVT roots, residual/total H and S, φ, f, vapour pressure and compressed-liquid fugacity	Z, V, density, fugacity, H, S and saturation pressure when roots permit
Numerical / plotting support	Newton-Raphson plus quadratic solution; bisection for T(P,V); matplotlib helper for plots	Cubic-root calculation in JavaScript; P-V graph on the page
Supporting files / interface	checks.py and plots.py; uses the math module for the main program	Gas selector, temperature and pressure inputs, root selector, plot selector and result tables
Documented operating units	K, bar, cm³/mol, J/mol and J/(mol K)	Inputs in K and bar; display includes density in kg/m³ and molar volume in cm³/mol

3.1 General calculation sequence

Irrespective of interface, the property calculation can be understood as a sequence of data retrieval, model parameterization, cubic solution, and thermodynamic post-processing. The diagram below is a new summary of that shared logic; it is not a screen capture from either program.

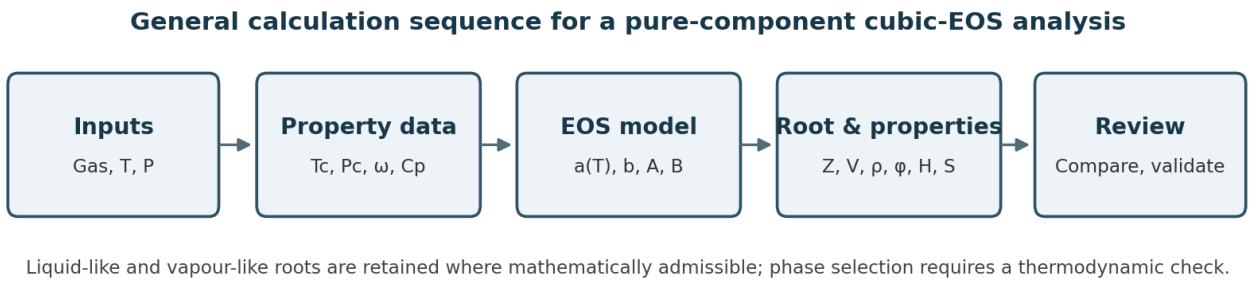


Figure 1. Generalized workflow for evaluating a pure gas with a cubic equation of state.

3.2 Numerical methods and user operation

- The Python program organizes critical properties and Cp coefficients in a gas dictionary, selects EOS constants, computes a(T) and b, solves for Z, and evaluates residual and ideal-gas contributions. Its menu provides six options after gas and EOS selection. The report gives the launch commands for the main, check and plotting scripts [1, pp. 3-4].
- The browser program asks the user to select a gas, enter positive temperature and pressure, and choose an automatic, vapour-like or liquid-like root. A separate selector controls the EOS used for the isotherm graph; the Calculate action refreshes the tables and graph [2, p. 2; p. 4].
- The browser graph is described as showing isotherms at reduced temperatures Tr = 0.8, 0.9, 1.0 and 1.2 together with the chosen state. The report provides an illustrative CO_2 P-V plot, but does not provide the raw coordinates used to draw its curves [2, p. 2-3].

3.3 Data handling and reproducibility

For reproducibility, a calculation record should contain the gas identity, Tc, Pc, ω, molar mass, Cp correlation, EOS choice, temperature, pressure, selected root, and the units used. Intermediate values A, B and all admissible Z roots should be retained so that the reported volume, density and fugacity can be checked. The two source reports do not provide identical gas databases or interfaces; therefore, a comparison should use the same gas and thermodynamic state wherever possible and should not assume that outputs from the two applications are directly interchangeable.

4. Reported results and numerical comparison

This section collects two different kinds of evidence from the supplied reports. The carbon-dioxide results are explicitly described by their source as representative calculator outputs that have not been experimentally validated [2, p. 3]. The n-butane data in the first report are comparisons with textbook benchmark values [1, p. 4]. Keeping these categories separate is essential when interpreting model accuracy.

4.1 Carbon dioxide example from the browser calculator

The second report gives a case at T = 300 K and P = 50 bar. The four EOS predictions are reproduced below without changing the reported digits [2, p. 3].

EOS	Z	V (cm³/mol)	ρ (kg/m³)	φ	f (bar)
vdW	0.7312	364.8	120.7	0.7924	39.621
RK	0.6971	347.8	126.6	0.7668	38.341
SRK	0.6943	346.4	127.1	0.7654	38.271
PR	0.6708	334.6	131.5	0.7464	37.320

For this state, every model gives Z below unity and φ below unity. Within this set of predictions, PR gives the lowest molar volume and highest density, while vdW gives the largest volume and lowest density. Since f = φP, all four reported fugacities fall below the applied pressure of 50 bar.

4.2 Independent arithmetic check using the PR row

A consistency check can be made from the tabulated PR volume without claiming a new EOS run. Using R = 83.14 $cm^3 \cdot bar \cdot mol^{-1} \cdot K^{-1}$, the ideal-gas molar volume at 300 K and 50 bar is:

$$V_{ideal} = RT/P = (83.14 \times 300)/50 = 498.84\ cm^3/mol$$

The tabulated PR volume then gives Z = V/Videal = 334.6/498.84 = 0.6708 (to four significant figures), consistent with the reported value. Using the CO_2 molar mass 44.01 g/mol also gives $\rho \approx 44.01/334.6 \times 1000 = 131.5\ kg/m^3$. These are arithmetic checks on the supplied table, not independent validation against measured data.

4.3 n-Butane textbook benchmark reported for the Python program

For n-butane at 500 K and 50 bar, the first report compares calculated Z, residual enthalpy and residual entropy with values tabulated in Smith et al. [1, p. 4].

EOS	Z: book	Z: program	H^R book (J/mol)	H^R program (J/mol)	S^R book (J/mol K)	S^R program (J/mol K)
vdW	0.6608	0.6610	-3937	-3935	-5.424	-5.420
RK	0.6850	0.6852	-4505	-4503	-6.546	-6.542
SRK	0.7222	0.7224	-4824	-4821	-7.413	-7.408
PR	0.6907	0.6909	-4988	-4985	-7.426	-7.421

Interpretation of provenance: this table is a source-reported reproduction of a textbook comparison. It is not a new result obtained by running the software in this report.

4.4 Comparison with reference equations of state

The Python report describes a separate comparison at 44 gas and supercritical states of methane, nitrogen, CO_2, propane, ethylene and ammonia over 10-200 bar, using reference values from high-accuracy equations of state. The reported mean absolute deviations are listed below [1, pp. 4-5].

EOS	Z deviation (%)	H^R deviation (J/mol)	S^R deviation (J/mol K)	φ deviation (%)
vdW	5.32	563	1.95	2.28
RK	1.81	162	0.53	0.80
SRK	2.68	66	0.20	1.01
PR	2.03	102	0.26	2.30

5. Discussion and engineering interpretation

5.1 Influence of the equation of state

A cubic EOS does not merely provide a different numerical route to the same answer: its attraction term, co-volume parameter and temperature correction alter the predicted compressibility and the derived properties. The CO_2 table illustrates this directly. For the selected state, Z ranges from 0.6708 with PR to 0.7312 with vdW, and the predicted molar volume spans 334.6-364.8 cm^3/mol [2, p. 3]. This spread is large enough to affect density and any downstream calculation that uses the molar volume.

The reference comparison in the Python report shows that no single metric identifies one universally best cubic model. RK has the smallest mean Z deviation in the stated gas/supercritical dataset (1.81%), whereas SRK has the smallest reported H^R, S^R and fugacity-coefficient deviations among the four models in that same table. The first report consequently recommends SRK and PR for many non-polar engineering applications, while noting that vdW errors are substantially larger in its comparison set [1, pp. 4-5]. The recommendation should be read within the tested compounds and conditions, not as a universal guarantee.

5.2 Meaning of the residual properties

Residual enthalpy and entropy quantify the correction required when an ideal-gas property estimate is transferred to a real-gas state. Their values depend on the EOS and the selected compressibility root. The negative residual H and S values in the n-butane benchmark indicate that, for those states and sign conventions, the real-gas properties lie below the corresponding ideal-gas reference contributions. Total H and S still require the ideal-gas heat-capacity path; reporting residual terms alone is not equivalent to reporting total properties [1, pp. 2, 4].

5.3 Fugacity as a phase-equilibrium criterion

The fugacity coefficient is not an arbitrary correction factor: it provides the non-ideal contribution needed for chemical-potential and phase-equilibrium calculations. For the CO_2 example, each reported φ is less than one, leading to $f < P$. For saturation calculations, the key condition is equality of liquid and vapour fugacities at a common temperature and pressure. The source reports that PR saturation pressures were within 0.3% of reference values for propane at 300 K, n-butane at 350 K and CO_2 at 280 K, and within 0.7% for ammonia at 300 K. Water at 373.15 K showed a 5.3% difference (0.960 versus 1.014 bar), attributed in the source to limitations of the standard PR alpha function for strongly polar, hydrogen-bonding fluids [1, p. 5].

5.4 P-V behaviour and root interpretation

The browser report's CO_2 isotherms illustrate the steep pressure increase at small molar volume as the excluded-volume term becomes important. Below the critical temperature, a cubic EOS may yield a non-monotonic section and multiple mathematical roots. The existence of multiple roots is a reminder that solving the polynomial is not the same as deciding which phase is stable. Root admissibility, fugacity equality and Gibbs-energy stability must be considered together [2, p. 3; 1, pp. 2-3].

5.5 Model choice in practice

- For non-polar gases, SRK and PR are reasonable starting points when enthalpy, entropy and phase behaviour are important; the choice should still be checked against fluid-specific reference data.
- A small Z error does not automatically imply small errors in H, S or fugacity. The metrics should be assessed separately for the property needed in the engineering task.
- Near the critical region, for liquid-volume predictions, and for polar or associating substances, a cubic model should be treated as an approximation and checked with a higher-accuracy property source.
- For a browser or command-line calculator, the user should retain the EOS label, root choice, units and reference conditions with every saved result.

6. Limitations, validation plan and conclusion

6.1 Limitations identified in the source material

- The Python report notes that cubic EOS impose fixed critical compressibility factors: 1/3 for RK/SRK and 0.30740 for PR. For n-butane, the source gives an experimental critical compressibility factor of 0.274, which helps explain limitations in liquid-volume predictions [1, p. 6].
- The first report notes reduced accuracy near critical conditions; for CO_2 at 320 K and 100 bar, PR gives Z = 0.390 compared with a reference value of 0.369 [1, p. 6].
- Strongly polar or associating fluids, including water, alcohols and ammonia, may require a fluid-specific alpha function or volume translation. The water saturation-pressure discrepancy is one example [1, pp. 5-6].
- The first report states that the Cp correlations are used over a finite temperature range, so absolute H and S values outside that range are extrapolations. The second report likewise warns that the stored correlations have a stated approximate range and that mixture calculations are not included [1, p. 6; 2, p. 4].
- The browser report describes its CO_2 table as representative outputs rather than experimentally validated values. Its saturation search may not converge if valid roots or a suitable pressure interval are unavailable [2, pp. 3-4].

6.2 Recommended validation checks

Before using a calculator for design or process decisions, the following checks should be completed and documented. These are recommended checks, not a claim that every one has already been run for the present report.

- Low-pressure limit: verify that Z approaches 1 and V approaches RT/P as pressure becomes sufficiently small.
- Unit consistency: verify conversions among bar, cm^3/mol, kg/m^3, J/mol and J/(mol K), including the molar-mass basis used in density.
- Root selection: verify that all retained roots satisfy the physical admissibility condition and that liquid/vapour selections follow the chosen stability rule.
- Phase equilibrium: below Tc, compare liquid and vapour fugacities at the calculated saturation pressure and test the convergence sensitivity to the pressure interval and initial guess.
- Reference comparison: test several states against a trusted thermodynamic database or published multiparameter equation of state, including cases near critical conditions and for polar fluids.
- Regression checks: preserve benchmark input-output cases so that code changes do not silently alter established results.

6.3 Conclusion

The two supplied projects demonstrate complementary ways to make cubic-EOS thermodynamics accessible: a compact Python program supports a broader 22-gas property database and detailed residual-property calculations, while a browser-based tool emphasizes guided inputs, result comparison and an interactive P-V view for nine gases. The common thermodynamic foundation links critical properties to the EOS parameters, solves a cubic equation for compressibility, and uses the selected root to estimate volumetric properties, residual H and S, and fugacity. The reported examples show that EOS choice can materially change calculated properties. They also show why a numerical answer should be accompanied by its provenance, phase-root choice and validation status. SRK and PR perform well for many of the non-polar cases tested in the Python report, but limitations remain near critical conditions and for polar or associating fluids. A defensible engineering workflow therefore combines a cubic EOS with transparent units, explicit root selection, reproducible benchmark tests and comparison against reliable reference data.

References

[1] “Cubic EOS Software: PVT, H, S and Fugacity of Real Gases,” supplied project report, 6 pages. Source material for the generalized cubic EOS, Python program structure, n-butane benchmarks, reference-data deviations, vapour-pressure checks and limitations.

[2] “Real Gas PVT Calculator: Thermodynamic Property Prediction Using Cubic Equations of State,” supplied project report, 4 pages. Source material for the browser calculator, gas list, CO_2 example at 300 K and 50 bar, P-V plot description, validation plan and limitations.

[3] J. M. Smith, H. C. Van Ness, M. M. Abbott and M. T. Swihart, Introduction to Chemical Engineering Thermodynamics, 8th ed., McGraw-Hill, 2018. Relevant material cited in [1]: Table 3.1; Sections 3.6, 6.4, 10.5, 13.6 and 13.7; Tables B.1 and C.1.

[4] I. H. Bell et al., “Pure and Pseudo-pure Fluid Thermophysical Property Evaluation and the Open-Source Thermophysical Property Library CoolProp,” Industrial & Engineering Chemistry Research, vol. 53, 2014, p. 2498 ff. Cited as reference-fluid data source in [1].

Source-use statement: numerical values identified as “reported” are transcribed from the two supplied reports. The CO_2 ideal-volume and density checks in Section 4.2 are arithmetic derived from the reported state and table. No experimental measurements, external validation, or fresh software execution are claimed in this document.